Formulating LPs — 1/3

The basic ingredients: cost function & constraints

$$\min: \ c^T x \qquad \left(u^T v = v^T u = \sum u_i v_i \right)$$

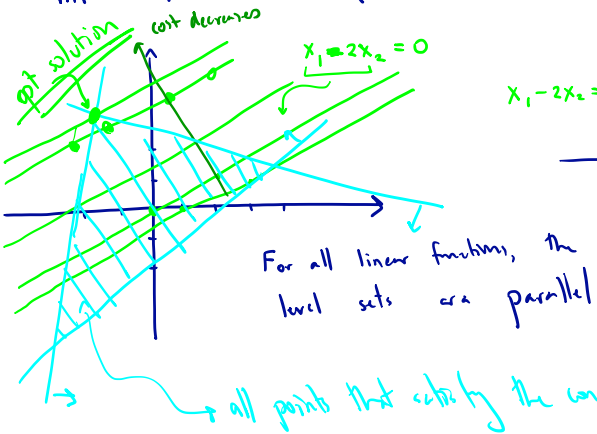
$$\begin{aligned}
\text{st:} \quad & a_1^T x = b \\
& \vdots \\
& a_m^T x = b_m \\
& a_{m+1}^T x \le b_{m+1} \\
& \vdots \\
& a_{m+k}^T x \le b_{m+k}
\end{aligned}$$

Formulating LPs — 2/3

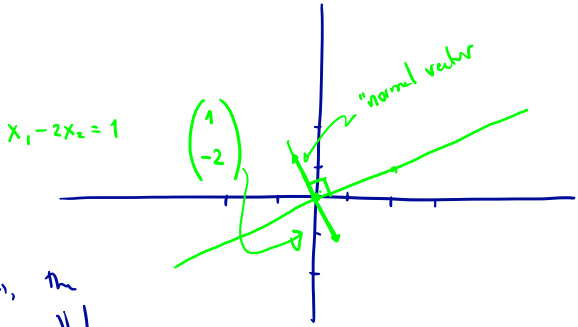
Cost function & level sets

$$\min: \ c^T x = \sum c_i x_i$$

$$\min: \ x_1 - 2x_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}^T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$



For all linear functions, the level sets are parallel



Formulating LPs — 3/3

Constraints — tight/binding/active constraints

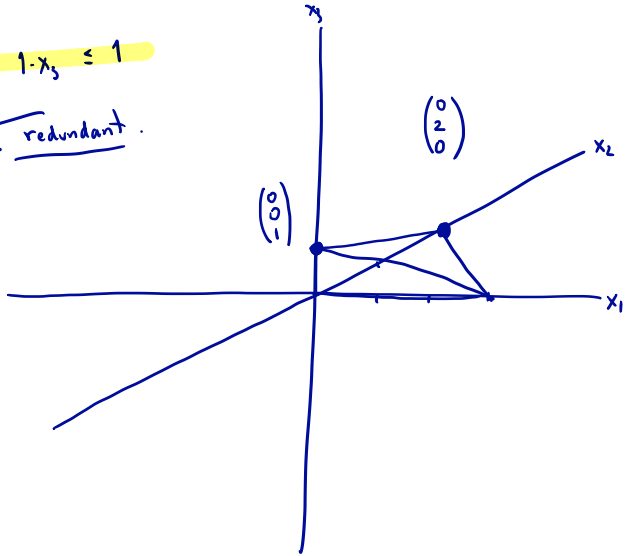
$$\frac{1}{3}x_1 + \frac{1}{2}x_2 + 1 \cdot x_3 \le 1$$

$$\cancel{x_1 + x_2 \ge 0}$$ redundant.

$$x_1 \ge 0$$

$$x_2 \ge 0$$

$$x_3 \ge 0$$



Example One

Production

n goods, good i sells for c_i per unit

m raw materials, b_i availability of raw material i.

objective: $\max: c^T x$

x_i = ~~# units~~ amount of product i

$$\text{st:} \quad \boxed{\begin{aligned} & a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \le b_1 \\ & \vdots \\ & a_{m1}x_1 + \cdots\cdots + a_{mn}x_n \le b_m \\ & x_1, \ldots, x_n \ge 0 \end{aligned}}$$

$$\begin{aligned} \max: \ & c^T x \\ \text{st:} \ & Ax \le b \\ & x \ge 0 \end{aligned}$$

a_{ij} = how much of material i is needed for one unit of product j

Example Two

Scheduling night nurses.

Hospital: night j $(j = 1, \ldots, 7)$, need d_j nurses on staff.

However: each nurse must work 5 days in a row.

Decision variables? Idea 1: let x_i = # nurses on day i

difficult to write down the 5-consecutive-day constraint.

Idea 2: x_i = # of nurses whose 5-day shift starts on day i.

min: $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7$ $\qquad x_i \geq 0$

st: $x_1 + \quad + x_4 + x_5 + x_6 + x_7 \geq d_1$ $\qquad x_i$ integer.

$x_1 + x_2 \quad + x_5 + x_6 + x_7 \geq d_2$

$\vdots$